

Jordan Nationwide 5G SA Network – Nominal Design Report

Scope: coverage, capacity, antenna/sector plan, frequency plan, fiber transport, 5G core and indoor coverage for a greenfield 5G Standalone network covering Jordan, with a **1 Gbps single-user downlink target in urban and suburban areas**. **Status:** nominal (pre-survey) design. Generated by `tools/generate_design.py` ; every number below is reproducible and changes when the inputs at the top of that script change.

Deliverable	File
Interactive satellite map (coverage, sites, sectors, fiber, indoor)	<code>index.html</code> (run <code>python -m http.server 8765</code> in this folder → <code>http://localhost:8765</code>)
Site list (6,270 sites: position, ground elevation, building height, antenna height, backhaul, hub, ring)	<code>output/sites.csv</code>
Cell list (18,675 sectors: azimuth, mechanical / digital / RET tilts, PCI, border flag)	<code>output/sectors.csv</code>
Fiber rings, hubs, indoor venues	<code>output/fiber_rings.csv</code> , <code>output/hubs.csv</code> , <code>output/indoor_venues.csv</code>
All KPIs, link budgets, compliance statistics	<code>output/summary.json</code>

1. Headline results

Item	Value
Macro sites / sectors	6,270 / 18,675 (DU 477 · U 2,970 · SU 2,245 · Rural 444 · Highway 134)
1 Gbps service footprint	1,491 km ² (all dense-urban, urban and suburban zones of 25 towns and cities)
Population inside designed zones	≈ 10.5 M of 11.3 M (93 %), plus highway corridors
Locations with ≥ 1 Gbps single-user DL at busy-hour load (outdoor, 300 MHz n78)	DU 97.9 % · U 97.6 % · SU 95.5 % (5th-percentile ≈ 1.0–1.1 Gbps, median ≈ 2.0 Gbps)
Same with 200 MHz / 100 MHz of n78	≈ 77–79 % / 25–33 % → 300 MHz of C-band is required for the guarantee
Busy-hour sector load (300 MHz)	DU 10.9 % · U 6.4 % · SU 3.8 % · Rural 23.5 %
Fiber route	5,737 km access + 887 km aggregation + 87 km metro core + 1,979 km backbone + 201 km spurs ≈ 8,900 km ; 5,910 sites (94 %) on fiber, 360 on microwave
Hubs / rings	150 C-RAN / pre-aggregation hubs, 849 access rings, 41 aggregation rings
Core	2 geo-redundant 5GC data centres (Amman W / E) + edge UPF in Amman Centre, Zarqa, Irbid, Aqaba; ≈ 2.5 Tbps busy-hour user plane
Indoor	51 venues, ≈ 5,340 n78 pRRUs + 196 n258 mmWave heads



Figure 1 – National design on satellite imagery: service-class zones, sites, DWDM backbone, core data centres / PoPs and indoor venues.

What “1 Gbps guaranteed when a user uses it” means in this design

A user who is alone on the air interface of a sector receives the whole carrier. His rate is then set only by bandwidth and SINR:

$$\text{rate} = \text{BW} \times 0.74 \text{ (TDD DL share)} \times 0.86 \text{ (1 - overhead)} \times \text{rank-adaptive spectral efficiency (} \leq 4 \text{ layers, 256-QAM, } 0.7 \times \text{Shannon)}$$

→ 1 Gbps needs SINR $\approx$ **13.9 dB with 300 MHz**, 17.7 dB with 200 MHz, 27.4 dB with 100 MHz. The grid is interference-limited (noise is irrelevant outdoors), so SINR is fixed by **neighbour-cell load**, not by power. The design therefore (a) gives every 1 Gbps sector 3×100 MHz of n78, and (b) dimensions the grid so busy-hour load stays ≈ 4 –11 %, where ≥ 95 % of locations reach 1 Gbps. Processor-sharing view: at 11 % load a user finds the sector idle ≈ 89 % of the time; when two users collide, MU-MIMO (64T64R, up to 16 layers) still serves both near 1 Gbps in mid-cell. **Expansion trigger:** busy-hour PRB utilisation > 20 % on a sector → add n258 layer or split. The residual 2–5 % of locations (deep wadis, cell corners) are covered by n258 hotspots, C-RAN coordinated scheduling / DL CoMP, or fall back to ≥ 700 Mbps.

Honest limits: this is a **median** prediction; deep indoor behind Jordanian stone façades reaches 1 Gbps in only 50–90 % of locations from the macro layer – that is why large buildings get dedicated indoor systems (§8) and homes are served by window/outdoor FWA CPE.

2. Inputs and method

Input	Source	Use
Terrain	Copernicus GLO-90 DEM (ESA, 3") – 29 tiles mosaicked	site ground elevation; nominals moved to the local high point in their search ring (4,806 sites moved, mean gain 17 m); per-sector terrain-aware tilt; knife-edge diffraction (ITU-R P.526) along every site→pixel profile; true 3-D antenna geometry in predictions; Dead Sea water mask
Building heights	GHSL GHS-BUILT-H R2023A (EU JRC, 100 m average net building height, epoch 2018)	antenna height = max roof within 250 m + 8 m clearance; rooftop vs monopole decision; 311 nominals deleted because their whole cell contains no buildings; verification of clutter classes
Borders / governorates	geoBoundaries ADM0 / ADM1	clipping, border-sector handling, statistics
Satellite imagery	Esri World Imagery (map backdrop)	digitising built-up zones, visual verification
Propagation	3GPP TR 38.901 UMa-NLOS (city), RMa-NLOS (rural) with LOS floor, + DEM diffraction	coverage / SINR / throughput rasters at 50 m (n78) and 300 m (n28)

Measured clutter statistics (GHSL at site locations): DU mean building 14.5 m (p95 27.7 m); U 8.1 m (p95 18.8 m); SU 3.5 m (p95 12.9 m).

Not yet included (needed before build): 3-D building vectors / 2 m clutter for ray-based prediction in DU, CW drive-test model calibration, candidate-site acquisition, structural surveys.

Spectrum actually held is an **assumption**. Public sources confirm only that TRC assigned parts of 3400–3800 MHz to Zain, Orange and Umniah for 5G (2022–23); per-operator amounts, 700 MHz and 26 GHz availability must be confirmed with TRC. A 300 MHz n78 holding implies a single wholesale / shared-RAN network or spectrum pooling.

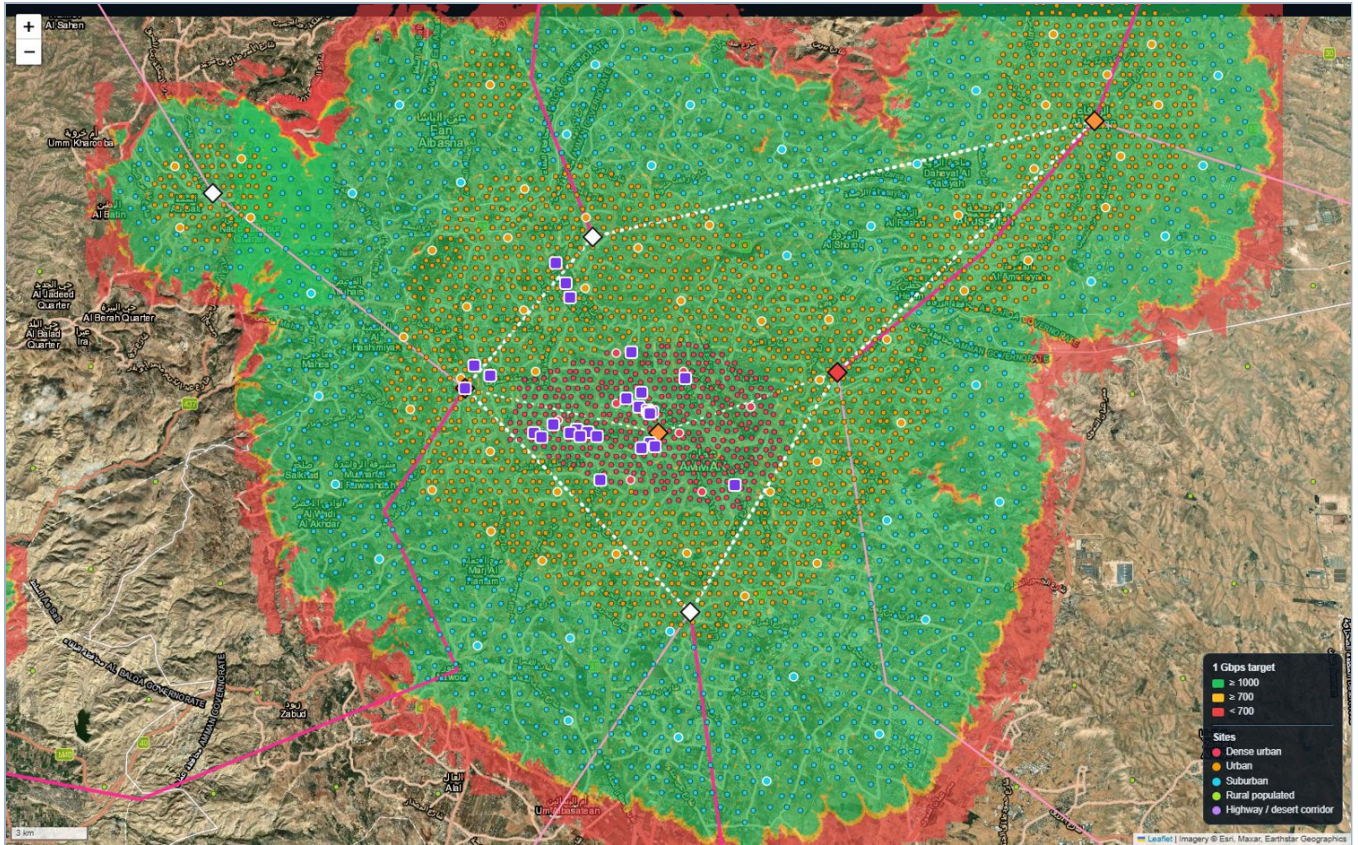


Figure 2 – Amman: terrain-aware single-user DL prediction (n78 300 MHz + FDD CA, 10 % neighbour load). Green ≥ 1 Gbps, amber 0.7–1 Gbps, red < 0.7 Gbps; wadis and cell corners stand out.

3. Frequency plan

Band	Spectrum	Duplex	Radio	Role
n28 (700 MHz)	2 × 10 MHz	FDD	2T4R city / 4T4R rural	National coverage layer, deep indoor, VoNR, UL anchor – on every site
n3 (1800 MHz)	2 × 20 MHz	FDD	4T4R	PCell + uplink for FDD-TDD CA (UL/DL decoupling), rural capacity
n1 (2100 MHz)	2 × 15 MHz	FDD	4T4R	Additional FDD capacity (DU, U, rural)
n78 (3400–3700 MHz)	3 × 100 MHz	TDD	64T64R (8T8R rural)	1 Gbps layer , reuse 1, 3-CC DL CA
n258 (26 GHz)	800 MHz (4 × 200)	TDD	street / indoor heads	DU hotspots, venues, FWA

- TDD frame: DDDSU, 30 kHz SCS, special slot 10:2:2, phase-synchronised nationally ($\pm 1.5 \mu\text{s}$) and coordinated with neighbouring-country 3.5 GHz networks and any legacy LTE-TDD in 3.5 GHz (fallback frame DDDDDDDSUU if required).
- 3700–3800 MHz left as guard / FSS earth-station protection and future 4th carrier.
- Carrier aggregation set: n3 (PCell, UL) + n1 + n78 × 3 $\rightarrow \approx 335$ MHz DL. UL on n78 where RSRP allows, otherwise on n3/n1.
- PCI: 336 groups × 3, greedy assignment with reuse distance $\geq 6 \times \text{ISD}$, mod-3 distinct within a site (values in `sectors.csv`). SSB: 8 beams (Case C) on carrier 1. PRACH: format B4 in cities, format 0 (14.5 km) rural/highway. TAC per hub cluster (~40 sites).

4. Radio design per service class

Class	Where	ISD	Cell range	Antenna height (result)	Sectors	Configuration
Dense urban (DU)	Amman core (downtown, Abdali, Shmeisani, Jabal Amman/Hussein, Sweifieh, east Amman), Irbid core – 58 km ²	375 m	250 m	25–49 m, mean 28 m	3	n78 64T64R 3×100 MHz + n1/n3 4T4R + n28 2T4R; n258 street cells in hotspots
Urban (U)	Amman, Zarqa, Russeifa, Irbid, Aqaba, Salt, Madaba, Mafraq, Jerash, Ajloun, Karak, Tafilah, Ma'an, Ramtha, Wadi Musa, Baqa'a – 534 km ²	450 m	300 m	30–36 m	3	as DU without n258
Suburban (SU)	outskirts of all the above, QAIA, Dead Sea hotel strip, Aqaba south coast/ASEZA, Jordan-Valley towns, Azraq, Mutah – 959 km ²	650 m	433 m	32–38 m	3	n78 64T64R 3×100 MHz + n3 4T4R + n28 2T4R
Rural populated	NW highlands and Karak–Tafilah plateau – 8,800 km ²	4.5 km	3 km	45 m tower	3	n28 4T4R + n1/n3 4T4R + n78 8T8R
Highway / desert	15 routes, ≈ 2,000 km	7 km	5 km	45 m tower	2 (along road)	n28 4T4R + n3 4T4R

4.1 Antenna system (per sector)

- n78 AAU** – 64T64R, 192 elements, 320 W, 400 MHz IBW; traffic-beam gain 24 dBi (SSB 20 dBi); horizontal scan ±60°; vertical HPBW ≈ 6°, digital tilt –2...+9°; ≈ 0.5 m², < 30 kg.
- Passive multiband antenna** 2L4H, 2.0 m (2.6 m rural): 2 ports 698–960 MHz (15.5 dBi, V-HPBW 9°) + 4 ports 1695–2690 MHz (17.5 dBi, V-HPBW 5.5°), independent RET 2–12° per band group; fed by n28 and n1/n3 RRUs mounted behind the antenna (jumper loss ≤ 0.5 dB).
- DU hotspots: n258 radios on street furniture at 8–10 m, 120–150 m apart.

4.2 Sectorisation and azimuth

- 3 sectors per macro site, azimuths 30° / 150° / 270°** on a hexagonal lattice (3GPP layout: each boresight points at a neighbour site's back-to-back null; ISD = 1.5 × cell range). All city sites share the pattern so the grid stays regular after the terrain micro-move.
- Highway sites: 2 sectors** aligned with the local road bearing (per-site values in `sectors.csv`).
- Rural/highway sectors whose footprint lies outside Jordan (border, Dead Sea, Gulf of Aqaba) are **removed**. City sectors whose boresight crosses a border or the sea within 2 km are kept, get **+2° tilt**, and are flagged for cross-border coordination: **109 sectors** (Aqaba–Eilat/Taba, Jordan Valley, Ramtha–Daraa, Dead Sea shore).
- Detailed planning should rotate individual sites ±20° to align with street canyons / avoid pointing along wadis; the regular plan is the starting point.

4.3 Tilt (terrain-aware, per sector)

total tilt = $\text{atan}((h_{\text{ant}} + z_{\text{site}} - \bar{z}_{\text{footprint}} - 1.5 \text{ m}) / \text{cell range}) + \text{V-HPBW}/2$, clamped to 2–14°.
 $\bar{z}_{\text{footprint}}$ = mean DEM height of 9 points at 0.5 / 0.75 / 1.0 × range and –30° / 0° / +30° off boresight. n78 takes up to 9° digitally; any remainder is mechanical (shared bracket), and mid-/low-band RETs are then set independently.

Class	Mean total n78 tilt	10th–90th percentile	Flat-terrain value
DU	9.6°	7–14°	8.4°
U	9.2°	6–13°	8.4°
SU	8.1°	5–12°	7.0°

Class	Mean total n78 tilt	10th–90th percentile	Flat-terrain value
Rural (low band)	4.6°	3–7°	5.3°
Highway (low band)	3.6°	3–4°	5.0°

The wide spread in Amman (hilltop sites looking into wadis need 12–14°, valley sites looking uphill 2–5°) is exactly why tilts must be terrain-driven. 1,190 sectors hit the 14° clamp – candidates for lower mounting height or site relocation in detailed design.

4.4 Antenna height (building-aware)

$h_{ant} = \max(\text{class default}, \max \text{ GHSL height within } 250 \text{ m} + 8 \text{ m})$, capped at 60 m. Rooftop if local buildings ≥ 9 m, else monopole. Result: 2,652 rooftops, 3,040 monopoles/greenfield, 578 lattice towers. GHSL is a 100 m average, so individual towers (Abdali, Jordan Gate, hotel towers) exceed it – these are handled by the indoor programme and by choosing the tallest roof in the search ring at acquisition.



Figure 3 – Amman core at street level: 375 m grid moved onto local high points, three sectors at 30° / 150° / 270°; squares are indoor venues.

5. Link budgets (per sub-carrier, 95 % area reliability)

Link	Class	EIRP/SC (dBm)	Req. SINR (dB)	Building loss (dB)	MAPL (dB)	Range (m)	Design range (m)
n78 300 MHz DL 1 Gbps	DU / U / SU	39.1	13.9	15 / 13 / 10	122.6 / 124.6 / 130.4	325 / 365 / 514	250 / 300 / 433 ✓
n78 200 MHz DL 1 Gbps	DU / U / SU	40.9	17.7	same	120.6 / 122.6 / 128.4	289 / 325 / 458	✓ (but fails on interference, §6)
n78 100 MHz DL 1 Gbps	DU / U / SU	43.9	27.4	same	114.0 / 116.0 / 121.8	194 / 218 / 308	✗

Link	Class	EIRP/SC (dBm)	Req. SINR (dB)	Building loss (dB)	MAPL (dB)	Range (m)	Design range (m)
n78 UL 20 Mbps (26 dBm UE)	DU / U / SU	-9.2	2.5	same	114.3 / 116.3 / 122.1	198 / 222 / 314	✗ at cell edge → UL moves to n3
n3 UL 10 Mbps (23 dBm UE)	DU / U / SU	-8.0	-0.8	12 / 10 / 7	119.2 / 121.2 / 127.0	373 / 420 / 592	✓
n28 DL 10 Mbps in-car	Rural / Hwy	36.0	4.4	8	141.4	5,280	3,000 / 5,000 ✓

Constants: UE NF 7 dB, gNB NF 3.5 dB, interference margin 3 dB (2 dB UL/low band), shadow margin 6.8 dB (σ 6 dB; SU 4.0 dB), noise -122.2 dBm per 30 kHz SC.

6. Capacity and 1 Gbps compliance

Traffic model (design year): 40 % of the population on this network (4.2 M subs), 80 GB/sub/month, 10 % of the day's traffic in the busy hour → 0.59 Mbps/sub, $\approx$ 2.5 Tbps nationally. Mean cell spectral efficiency 10 b/s/Hz (64T64R MU-MIMO), 4 b/s/Hz for rural 8T8R.

Class	Pop. density	Subs / sector	Offered BH Mbps / sector	Load @100 MHz	@200 MHz	@300 MHz
DU	25,000 /km ²	407	241	32.6 %	16.3 %	10.9 %
U	10,000 /km ²	240	142	19.2 %	9.6 %	6.4 %
SU	2,500 /km ²	142	84	11.4 %	5.7 %	3.8 %
Rural	150 /km ²	396	235	57.5 %	33.3 %	23.5 %

Terrain-aware compliance – % of 50 m pixels with single-user DL $\geq$ 1 Gbps (n78 + 35 MHz FDD CA), outdoor:

Neighbour load	DU 100 / 200 / 300 MHz	U 100 / 200 / 300 MHz	SU 100 / 200 / 300 MHz
10 % (design busy hour)	25 / 77 / 97.9	28 / 79 / 97.6	33 / 78 / 95.5
20 % (expansion trigger)	11 / 51 / 78	14 / 54 / 80	18 / 55 / 80
30 %	6 / 37 / 62	8 / 40 / 65	12 / 44 / 66
50 %	2 / 22 / 44	3 / 25 / 47	6 / 30 / 50

At the design point (300 MHz, 10 %): in-car/light indoor 97.5 / 96.5 / 88.2 %; deep indoor 90.4 / 82.8 / 50.4 %. SS-RSRP $\geq$ -105 dBm over 100 / 99 / 96 % of DU / U / SU area.

Conclusions 1. The site grid is sufficient; **bandwidth and load control deliver the guarantee**. 100 MHz cannot do it at any density. 2. Keep busy-hour load under $\approx$ 15 %: QoS flow with GBR for the 1 Gbps tariff, fair-use shaping of bulk traffic, n258 offload where utilisation > 20 %. 3. C-RAN in DU/U (\$7) enables coordinated beamforming / CoMP, worth $\approx$ 3–5 dB at cell edges – the margin that lifts the last 2–5 % of pixels.



Figure 4 – n28 (700 MHz) national coverage layer with DEM diffraction: populated west, highway and desert corridors.

7. Transport, fiber and core

RAN architecture. DU/U: **C-RAN** – AAUs connect by eCPRI (7-2x), $3 \times 25\text{GE}$ per AAU over dark fiber ($\leq 10\text{ km}$, $\leq 75\text{ }\mu\text{s}$) to DU hotels co-located with the 150 hubs. SU/rural: D-RAN with DU at site, 25GE backhaul.

Tier	Topology	Capacity	Fiber
Access	849 rings, ≤ 8 sites, dual-homed to hub; ring geometry generated per hub (angular sectors, out-and-back)	25GE (50GE-ready); site peak = 5.2 Gbps sector peak + mean of other sectors $\approx 8\text{ Gbps}$	48-core, 5,737 km route (street factor 1.35); 12-core site drop
Aggregation	41 rings of ≤ 5 hubs to a PoP	100GE, SRv6 + EVPN, FlexE hard slices	96-core, 887 km
Metro core	Amman ring DC1 – North – Zarqa – DC2 – South – DC1, Centre PoP dual-homed	$N \times 400\text{GE}$	87 km
Backbone	Southern national ring: Amman – Desert Highway – Ma'an – Aqaba – Wadi Araba – Dead Sea – Amman. Northern ring: Amman – Zarqa – Mafrq – Ramtha – Irbid – Jerash – Amman. Regional spurs: King's Highway (Madaba–Karak–Tafilah–Petra–Ma'an), Jordan Valley, Salt–Deir Alla, Ajloun, Karak–Qatraneh, Tafilah–Jurf, Azraq, Safawi–Ruwaished, Wadi Rum	DWDM/OTN $N \times 400\text{G}$ (national), 100G (regional)	1,979 km; doubles as highway-site backhaul (add/drop $\leq 2\text{ km}$, 201 km of spurs)
Rural off-route	360 sites	E-band $10\text{G} \leq 6\text{ km}$, else 6–18 GHz XPIC	–

Synchronisation: GNSS + ePRTC at DCs/PoPs, PTP G.8275.1 with full on-path support and SyncE; $\pm 1.5\text{ }\mu\text{s}$ network-wide, $\pm 130\text{ ns}$ inside C-RAN clusters for CA/CoMP; GNSS at every hub as backup.

5G core (SA, cloud-native, 3GPP Rel-17): AMF, SMF, AUSF, UDM/UDR, PCF, NRF, NSSF, NEF, SCP, BSF, CHF and IMS (VoNR – there is no LTE fallback in a pure-SA network, so VoNR on n28/n3 is mandatory from day 1) active-

active in **DC1 Amman West** and **DC2 Amman East**; UDR replica / DR in Irbid. **Distributed UPF + MEC + CDN:** DC1/DC2 $\approx$ 55 %, Amman Centre + Zarqa 15 %, Irbid 15 %, Aqaba 8 % (also international gateway at the submarine-cable landing), remainder regional. N+1 UPF pools sized for 2.5 Tbps. Slices: eMBB-1G (GFBR), FWA, enterprise/private, public safety, O&M.

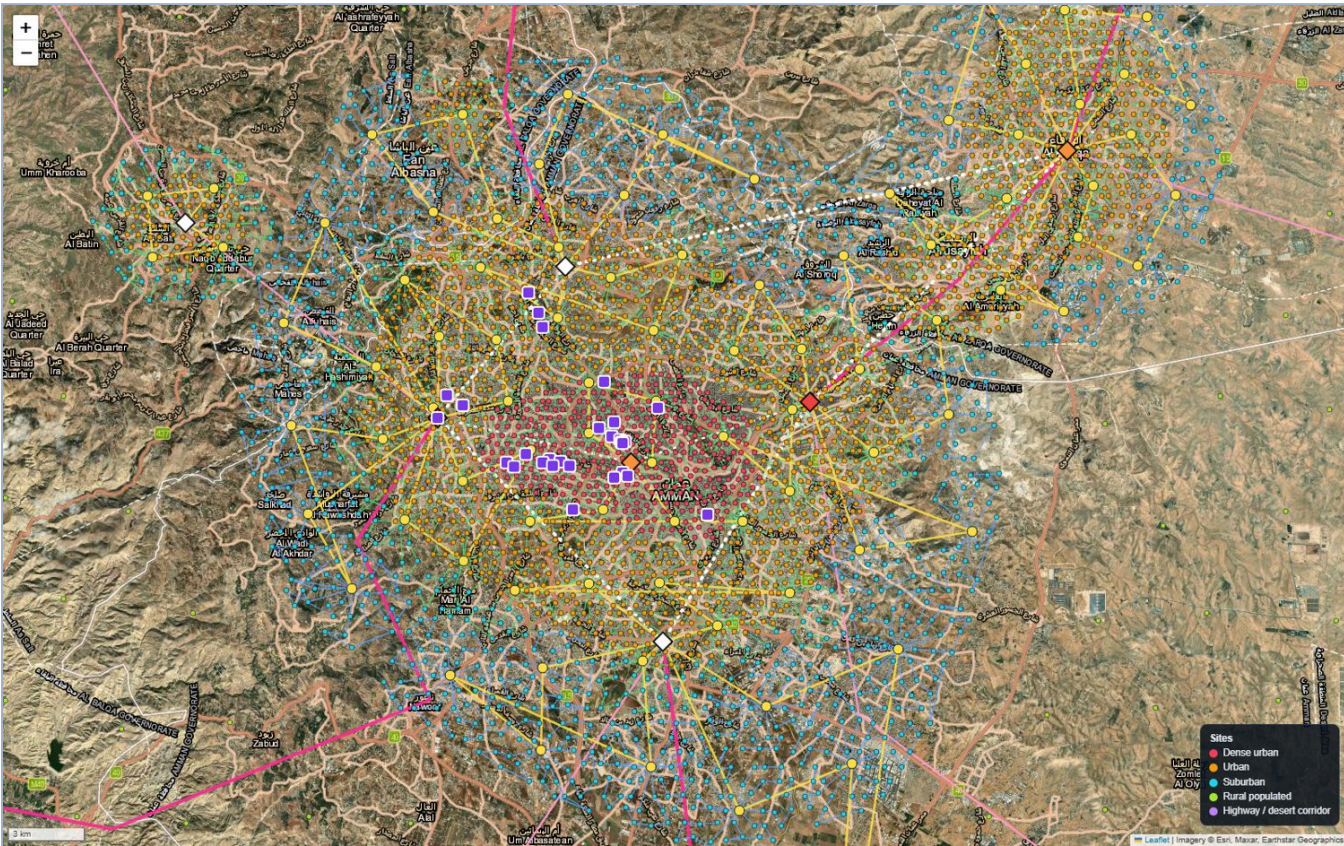


Figure 5 – Amman transport: C-RAN access rings (green), D-RAN rings (blue), 100GE aggregation rings (yellow), hubs, metro core (white) and backbone (magenta).

8. Indoor coverage (malls, hotels, airports, hospitals, campuses)

Macro n78 loses 20–30 dB through Jordanian stone/concrete façades, so every large building gets a dedicated **digital indoor system** (passive DAS cannot carry 4×4 MIMO at 3.5 GHz economically).

Venue type	Design rule	Example result
Mall / mixed-use	n78 4T4R pRRU (3×100 MHz, 4 × 250 mW) per $\approx$ 550 m ² of public area, 20–25 m grid, CAT6A/hybrid-fiber to RHUB (8 pRRU each); n258 heads in atria and food courts	Abdali Mall 248 pRRU + 10 mmW; Mecca Mall 208; City Mall 175
Hotel / resort	1 pRRU per $\approx$ 5 rooms (corridor-mounted), + public areas per 500 m ² , ballroom/lobby mmWave	Amman Rotana 125 pRRU; Tala Bay cluster 252
Airport / convention	per 600 m ² + mmWave at gates, check-in, halls	QAIA 172 pRRU + 13 mmW
Hospital / campus	per 650 m ² of served floor, EMC review in clinical areas	University of Jordan 270
Stadium	n78 + n258 narrow-beam bowl sectors ($\approx$ 1 per 700 seats)	Amman Intl Stadium 24 mmW sectors

Indoor budget: 24 dBm/port, InH-NLOS 76 dB at 15 m → RSRP $\approx$ −85 dBm, SINR > 25 dB, rank 4 → 1.5–3 Gbps per user. Dedicated PCIs, reduced SSB power at entrances, 3 dB HO offset to stop leakage; each venue dual-fiber homed (2 × 10GE, 2 × 25GE above 60 pRRUs) to the nearest hub; in-building DU. Total for the 51 listed venues

(Amman, Dead Sea, Aqaba, Petra, Irbid): $\approx 5,340$ pRRUs, 196 mmWave heads. Venue coordinates and areas are approximate – confirm by survey; phase 2 adds every building $> 10,000\text{ m}^2$ or > 8 floors.

9. Sites by governorate

Amman 2,455 · Zarqa 883 · Irbid 857 · Balqa 510 · Mafraq 285 · Aqaba 280 · Karak 257 · Ma'an 219 · Madaba 173 · Jerash 133 · Tafilah 115 · Ajloun 103.

10. Risks and next steps

1. **Spectrum** – secure $3 \times 100\text{ MHz}$ n78 (or accept $\approx 78\%$ compliance with 200 MHz and fill with n258). Confirm 700 MHz and 26 GHz availability with TRC.
2. **Model calibration** – CW measurements in Amman (hills + stone clutter), Irbid, Aqaba; replace UMa with a tuned SPM/ray model on 2–5 m building data for DU.
3. **Site acquisition** – 6,270 nominals with $0.2 \times \text{ISD}$ search rings; reuse of existing operator towers/rooftops should be maximised (sharing can cut new builds by $> 50\%$).
4. **Cross-border coordination** – 109 flagged sectors; TDD frame alignment with neighbours.
5. **Phasing** – P1 Amman–Zarqa–Irbid–Aqaba DU/U + backbone ($\approx 3,400$ sites); P2 suburban + governorate capitals; P3 rural/highway + indoor phase 2.
6. **Device ecosystem** – 1 Gbps needs UEs supporting n78 3-CC DL CA (300 MHz) with 4 Rx; verify the target handset/CPE portfolio.

11. Deployment summary – 10 rollouts over 36 months

Rollout	On-air window	Sites	DU	U	SU	Rural	Hwy	Hubs	Access + agg fiber km	Backbone km	Indoor venues	Cum. sites	Cum. pop. covered	Cum. 1 Gbps km ²	Main areas
RO-01	Apr 2027 – Jun 2027	306	293	13	0	0	0	6	260	87	0	306	8.6 %	40	Amman-Zarqa Metro
RO-02	Jul 2027 – Sep 2027	465	176	289	0	0	0	10	445	194	18	771	18.4 %	117	Amman-Zarqa Metro, Irbid
RO-03	Oct 2027 – Dec 2027	617	8	589	20	0	0	15	645	340	5	1,388	28.5 %	236	Amman-Zarqa Metro, Aqaba, Irbid
RO-04	Jan 2028 – Mar 2028	719	0	625	61	0	33	16	719	337	14	2,107	39.3 %	376	Amman-Zarqa Metro, Irbid, Aqaba, Desert Highway (R15) Amman-Aqaba

Rollout	On-air window	Sites	DU	U	SU	Rural	Hwy	Hubs	Access + agg fiber km	Backbone km	Indoor venues	Cum. sites	Cum. pop. covered	Cum. 1 Gbps km ²	Main areas
RO-05	Apr 2028 – Jun 2028	769	0	561	176	0	32	19	806	278	1	2,876	50.1 %	550	Amman-Zarqa Metro, Wadi Musa / Petra, Irbid, Dead Sea - Wadi Araba (R65) Amman-Aqaba
RO-06	Jul 2028 – Sep 2028	770	0	455	308	0	7	19	916	174	9	3,646	60.3 %	755	Irbid, Amman-Zarqa Metro, Madaba, Mafraq
RO-07	Oct 2028 – Dec 2028	766	0	264	191	311	0	13	516	185	0	4,412	73.9 %	878	Jerash, Ma'an, Salt, Mafraq
RO-08	Jan 2029 – Mar 2029	718	0	166	415	126	11	18	774	471	2	5,130	83.4 %	1,071	Amman-Zarqa Metro, Ajloun, Amman governorate (rural), Salt
RO-09	Apr 2029 – Jun 2029	669	0	2	609	7	51	17	885	0	0	5,799	88.9 %	1,309	Amman-Zarqa Metro, Irbid, Mafraq-Safawi-Ruwaished (R10), Zarqa-Azraq-Safawi
RO-10	Jul 2029 – Sep 2029	471	0	6	465	0	0	17	659	0	2	6,270	93.0 %	1,491	Amman-Zarqa Metro, Irbid, Mutah-Mazar, Aqaba

Cumulative population = share of Jordan's population living inside the footprint built so far (design-model estimate).

Year 1 (2027, RO-01...RO-03): 1,388 sites. Core live in M3–M4; all of dense-urban Amman and Irbid core, inner urban Amman, Aqaba city; northern backbone ring and Desert Highway lit. ≈ 28.5 % of population covered. **Year 2 (2028, RO-04...RO-07): 3,024 sites.** Remaining urban Amman, Zarqa, Russeifa, Irbid; most governorate capitals (Ajloun and the last Salt/Madaba clusters follow in RO-08); Dead Sea and Petra; southern backbone ring closed; national highways; first rural packages. ≈ 73.9 %. **Year 3 (2029, RO-08...RO-10): 1,858 sites.** Suburban rings,

remaining rural, eastern desert corridors, n258 hotspot layer in dense urban, indoor phase completion, network-wide optimisation and hand-over. 93.0 %.

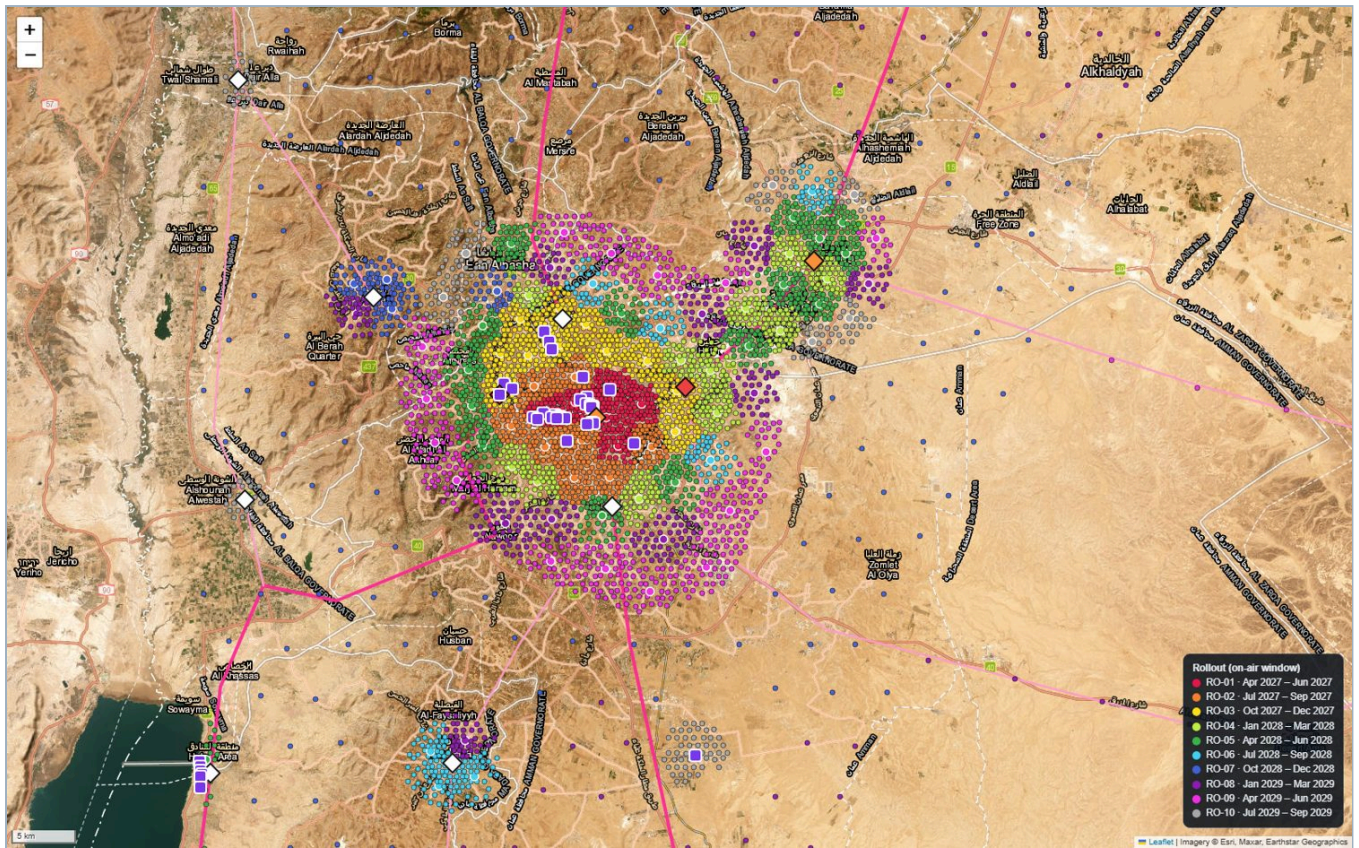


Figure 6 – Amman-Zarqa rollout sequence: inside-out from the dense-urban core (RO-01) to the suburban ring (RO-09/10). Full plan in the deployment/ folder.

Data credits: Copernicus DEM © ESA/DLR/Airbus; GHSL © European Commission JRC (CC BY 4.0); geoBoundaries (CC BY); imagery © Esri and its providers.